

1 An algorithm for the maximum flow problem

Now consider the problem of sending material in an undirected graph $G = (V, E)$ from a given *source node* to a given *sink node*. In other words, material flows from a source node along the edges of a graph to a sink node. This means that the material simply flows through nodes in the graph that are *not* the source node and *not* the sink node.

Every edge of the network, or the graph, has two numbers associated with it. For an edge $(u, v) \in E$, one number states the *capacity* available for sending flow in the direction from u to v . The other number states the capacity available for sending flow in the direction from v to u . The question is: How much flow can be send from the source node to the sink node? This problem is also called the *maximum flow* problem. We now present, by example, a method for solving maximum flow problems.

The graph in Fig. 3 gives the data for a maximum flow problem, where node 1 is the source node and node 7 is the sink node. For the edge $(1, 2)$, the number 5 states that at most 5 units can flow from node 1 to node 2, whereas nothing (zero) can flow in the opposite direction.

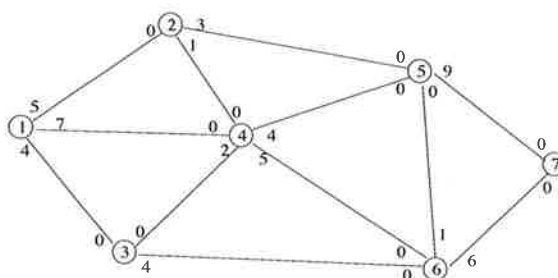


Figure 1: Data for a maximum flow problem

The algorithm we present first builds a tree containing all nodes that can be reached from the source node with a path with positive capacity. Nodes that can be reached directly from the source node (with an edge with positive capacity) are assigned distance level 1. Nodes that are *not* at distance level 1, but can be reached from a node at distance level 1 (with an edge with positive capacity) are assigned distance level 2. This is continued until no more nodes can be reached. Fig. 2 below gives the graph obtained by labeling the nodes of Fig. 1 in this way. The labeling algorithm is called *breadth-first-search*, and the resulting tree is called a *breadth-first tree*.

As can be seen in Fig. 2, there is a path with positive capacity from the source node to the sink node, namely the path $1 \rightarrow 2 \rightarrow 5 \rightarrow 7$. The amount of flow that can flow along this path is $f = 3$ (the capacity of the edge from node 2 to node 5).

We now adjust the capacities of the original network to reflect this flow. We would like the resulting *residual graph* to represent the amount of *additional* flow that can flow along each edge. For an edge where no flow is flowing, the capacity in the residual graph is the same as the original capacity. For example, for the edge $1 \rightarrow 2$, a flow of 3 is now flowing, so its residual (remaining)

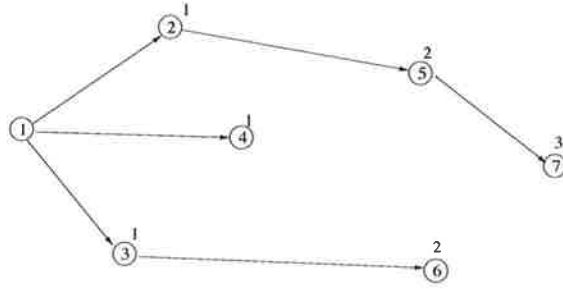


Figure 2: Breath-first tree based on the initial capacity data

capacity is given by $5 - 3 = 2$. For the opposite edge $2 \rightarrow 1$, there was a capacity of zero, and there is now a possibility of cancelling the flow of 3 on the edge $1 \rightarrow 2$. The residual capacity of the edge $2 \rightarrow 1$ is therefore given by 3. Fig. 3 gives the residual graph.

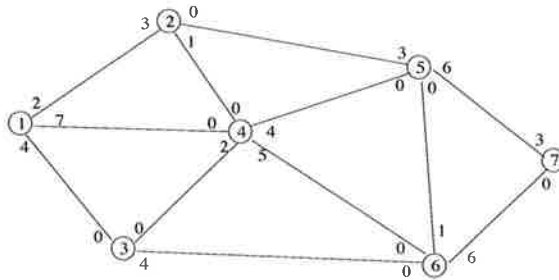


Figure 3: Residual graph after obtaining an initial flow of 3 units

The next step of the algorithm is to decide whether additional flow can flow from the source node to the sink node (more flow than is currently flowing). To decide this, we use the residual graph and check whether the sink node can be reached from the source node with a path with positive residual capacity. To obtain such a path, we build a breath-first tree that only uses edges with positive residual capacity. This tree is constructed from the residual graph of Fig. 3 in the same way as was done initially, and the resulting breath-first-tree is shown in Fig. 4. Note that the breath first tree may not be unique. This is because there might be several possibilities for reaching a node at distance $(k + 1)$ from a node at distance k . Nevertheless, any node that *can* be reached with a path of positive capacity will be reached by a path in the breath first tree.

From Fig. 4 we see that there is the path $1 \rightarrow 4 \rightarrow 6 \rightarrow 7$ from the source node to the sink node which has a positive residual capacity. The residual capacity of this path is 5, namely the residual capacity on the edge $4 \rightarrow 6$. We are therefore able to send an *additional* flow of 5 units from the source node to the sink node.

The total flow that is now flowing after adding the flow obtained from the two *augmenting paths* $1 \rightarrow 2 \rightarrow 5 \rightarrow 7$ and $1 \rightarrow 4 \rightarrow 6 \rightarrow 7$ is given in Fig. 5. We have now managed to send a total flow of $5 + 3 = 8$ units from the source node to the sink node.

We next update the residual capacities of Fig. 3 to reflect the total amount of flow that is now flowing. The resulting residual graph is shown in Fig. 6.

The next step is then to decide whether further flow can be send from the source node to the sink node. To do this, we build a breath-first tree based on the residual graph of Fig. 6. The breath-first tree that is obtained is shown in Fig. 7.

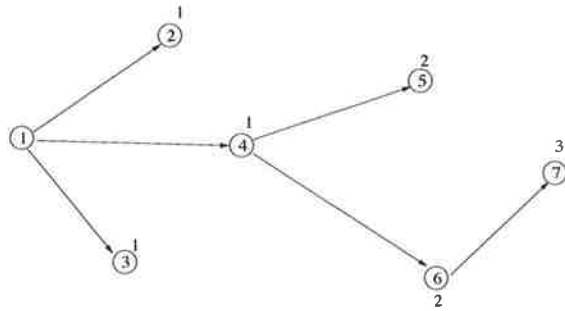


Figure 4: Breath-first tree from the first residual graph

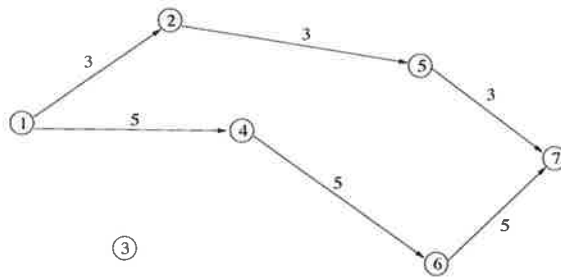


Figure 5: Total amount flowing after two augmenting paths

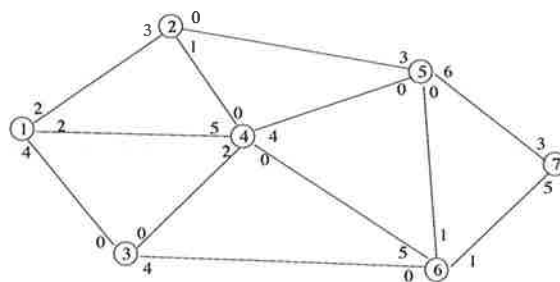


Figure 6: Residual graph after two augmenting paths

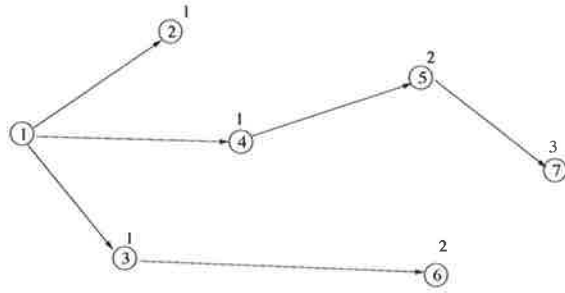


Figure 7: Breath-first tree after two augmenting paths

From the breath-first tree of Fig. 7 we see that there is the path $1 \rightarrow 4 \rightarrow 5 \rightarrow 7$ with capacity 2 from the source node to the sink node. We therefore update the current flow to reflect this additional flow. The graph in Fig. 8 gives the total flow that is flowing after three augmenting paths have been identified (edges with a flow of zero are not shown). The total flow that is now flowing from the source node to the sink node is given by $3 + 5 + 2 = 10$.

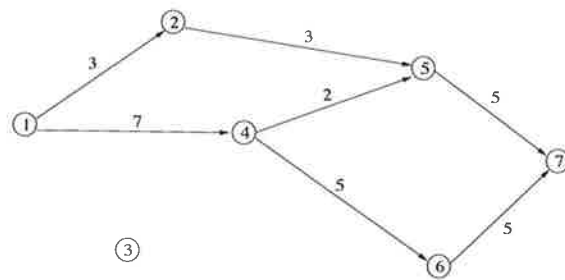


Figure 8: Total amount flowing after three augmenting paths

Again we need to update the residual capacities of Fig. 6 to reflect the additional flow on the third path. The resulting residual graph is shown in Fig. 9.

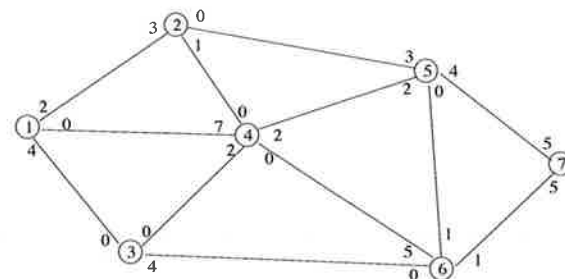


Figure 9: Residual graph after three augmenting paths

We next construct a breath-first tree based on the residual graph of Fig. 9. We obtain the breath-first tree in Fig. 10. Observe that node 4 is now no longer at distance 1, since there is no further residual capacity on the edge from the source node to node 4.

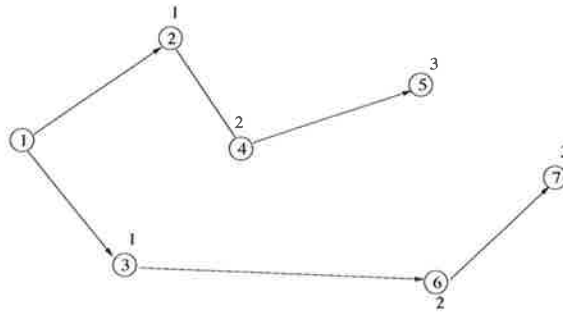


Figure 10: Breath-first tree from the residual graph of Fig. 9

From Fig. 10 we see that the residual graph of Fig. 9 contains the path $1 \rightarrow 3 \rightarrow 6 \rightarrow 7$ with a positive residual capacity of 1. This then gives a new augmenting path. Updating the graph of Fig. 8 by adding the additional flow of 1 unit from the fourth augmenting path then gives the total flow in Fig. 11. Observe that we have now managed to send a total flow of $3 + 5 + 2 + 1 = 11$ units from the source node to the sink node.

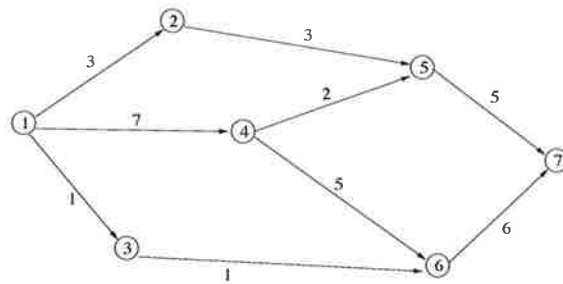


Figure 11: Total amount flowing after four augmenting paths

We next update the residual graph of Fig. 9 to reflect the extra flow on the fourth path. The resulting residual graph (after having identified four augmenting paths) is given in Fig. 12.

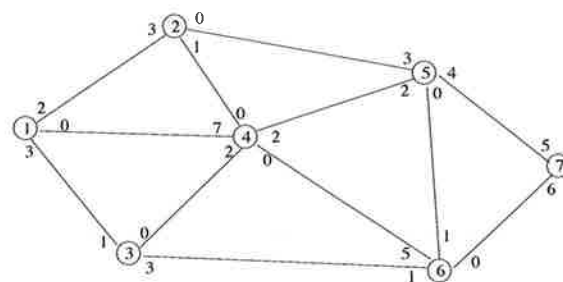


Figure 12: Residual graph after four augmenting paths

Applying the breath-first search method to the graph of Fig. 12 then gives the breath-first tree of Fig. 13.

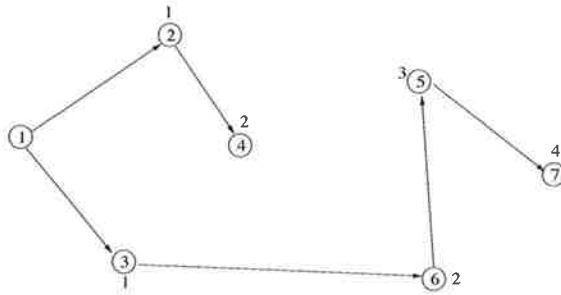


Figure 13: Breath-first tree from the residual graph of Fig. 12

From Fig. 13 we see that the residual graph of Fig. 12 contains the path $1 \rightarrow 3 \rightarrow 6 \rightarrow 5 \rightarrow 7$ from the source node to the sink node with a residual capacity of 1 unit. Updating the flow of Fig. 11 to take an extra flow of 1 unit from the fifth augmenting path into account gives the total flow that is shown in Fig. 14. We now have a total flow of $3 + 5 + 2 + 1 + 1 = 12$ units flowing from the source node to the sink node.

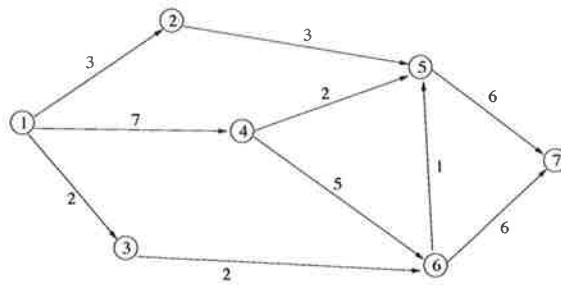


Figure 14: Total amount flowing after five augmenting paths

We next update the residual graph of Fig. 12 to reflect the new flow along the fifth path. The resulting residual graph is given in Fig. 15.

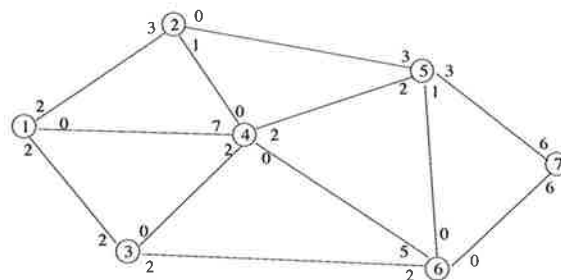


Figure 15: Residual graph after five augmenting paths

The next step is to construct a breath-first tree from the residual graph of Fig. 15. The tree obtained is given in Fig. 16.

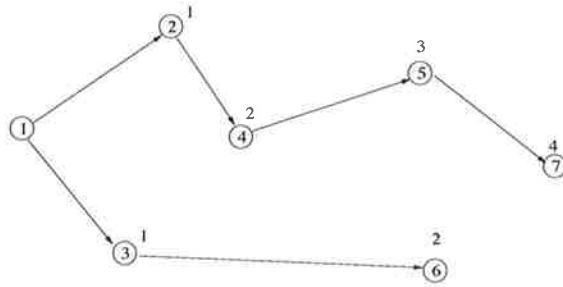


Figure 16: Breath-first tree from the residual graph of Fig. 15

Fig. 16 shows that there is an augmenting path given by $1 \rightarrow 2 \rightarrow 4 \rightarrow 5 \rightarrow 7$ from the source node to the sink node of capacity 1 unit. Updating the flow of Fig. 14 to take this new flow into account gives a total flow of $3 + 5 + 2 + 1 + 1 + 1 = 13$ as shown in Fig. 17.

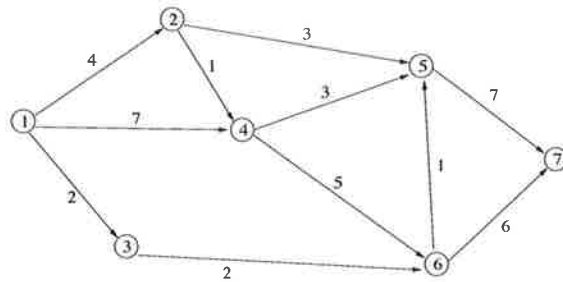


Figure 17: Total amount flowing after six augmenting paths

We next update the residual graph of Fig. 15 to take the extra flow of 1 unit along the sixth augmenting path into account. The resulting residual graph is given in Fig. 18.

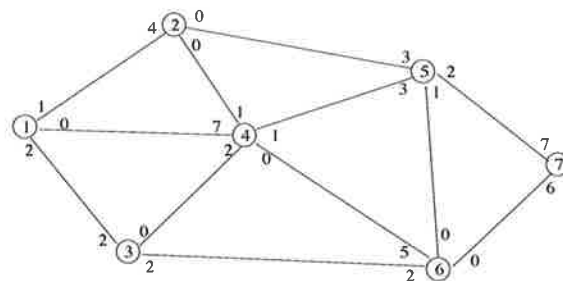


Figure 18: Residual graph after six augmenting paths

The next step is then to construct a breath-first tree corresponding to the residual graph of Fig. 18. This tree is shown in Fig. 19.

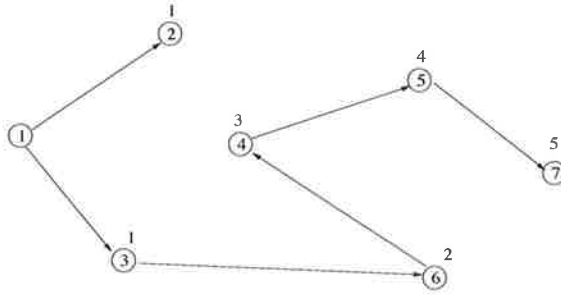


Figure 19: Breath-first tree from the residual graph of Fig. 18

From Fig. 19 we see that there is an augmenting path on which it is possible to send additional flow. This path is given by $1 \rightarrow 3 \rightarrow 6 \rightarrow 4 \rightarrow 5 \rightarrow 7$ and has a residual capacity of 1 unit. Sending one unit of extra flow along this path, and updating the flow of Fig. 17 correspondingly, results in the total flow depicted in Fig. 20. This flow has cardinality $5 + 3 + 2 + 1 + 1 + 1 + 1 = 14$.

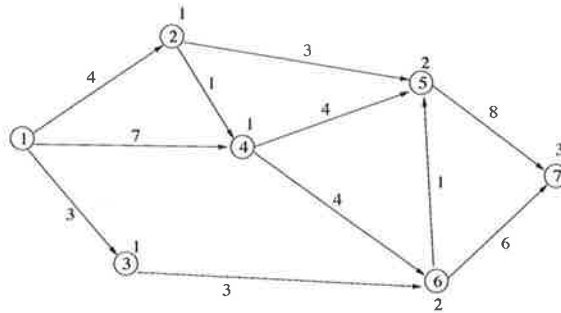


Figure 20: Total amount flowing after seven augmenting paths

Observe that, in order to update the graph of Fig. 17 to take the flow of the seventh augmenting path into account, it was necessary to cancel previous flow along the edge $4 \rightarrow 6$. We now update the residual graph of Fig. 18 to take the flow along the seventh augmenting path into account. This gives the residual graph in Fig. 21.

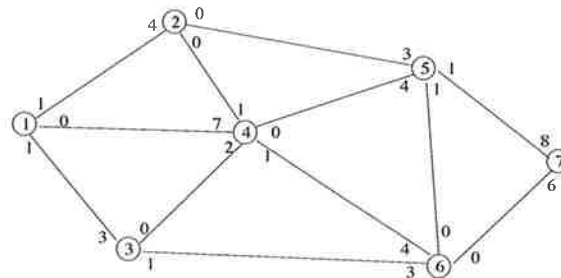


Figure 21: Residual graph after seven augmenting paths

From the residual graph of Fig. 21, we now construct a breath-first tree. This tree is given in Fig. 22. As can be seen from this tree, there is no longer an augmenting path from the source node to the sink node. This means that the current flow from the source to the sink (given in Fig. 20) is as large as possible.

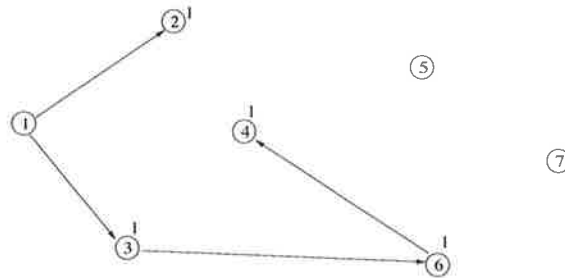


Figure 22: Breath-first tree from the residual graph of Fig. 23

It can be shown that a flow from a source to a sink is as large as possible, if and only if the corresponding residual graph does not have an augmenting path. We will not prove this result here.